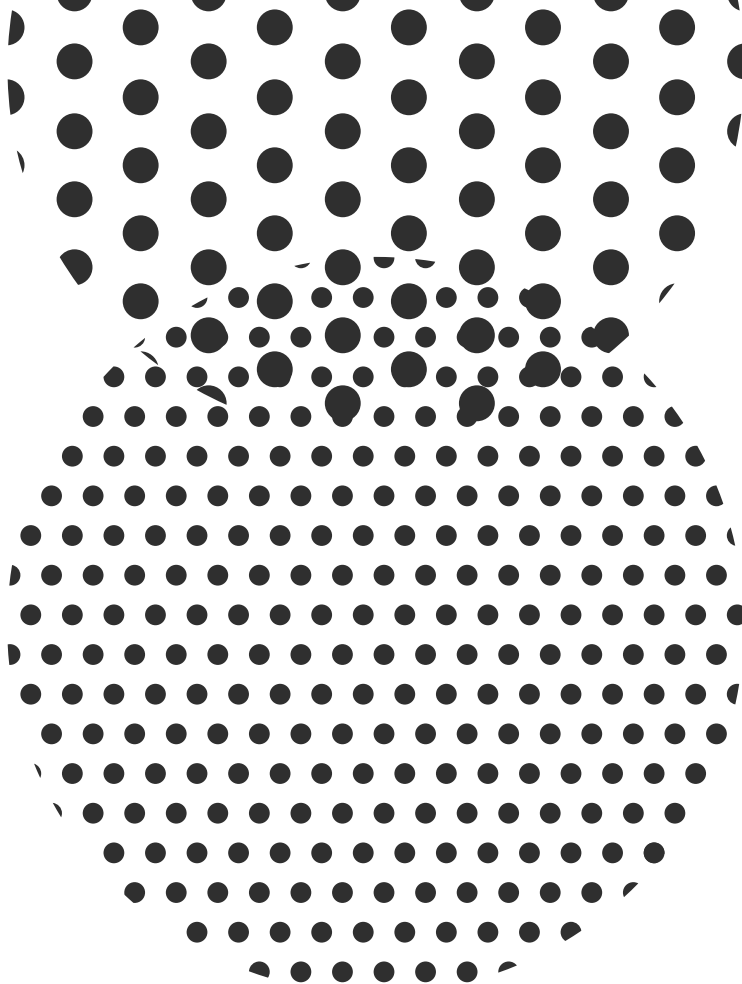




1 How does the 4-dimensional regular representation of A_4 reduce?

Any finite group G has $N(G)$ -dimensional representation, known as the regular rep..

$\Rightarrow A_4$ has 4-dimensional representation

Let's start by writing character of A_4 for 4-d. representation to the size of the class table:

A_4	n_c		1	1'	1''	3	4
	1	I	1	1	1	3	4
Z_2	3	(12)(34)	1	1	1	-1	0
Z_3	4	(123)	1	ω	ω^*	0	1
Z_3	4	(132)	1	ω^*	ω	0	1

For 4-d rep the id has $\chi(I) = d = 4$

in 4-d rep (12)(34) can be written as

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \Rightarrow \chi = 0$$

For (123) & (132) we have tr 1 since

$$(123) = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \text{tr} = 1$$

the 3x3 matrix that remains when removing final row & column is (123) in 3-d rep for A_3

$$(132) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \text{tr} = 1$$

Let's now use column orthogonality

$$\sum \text{Col}_4 \cdot \text{Col}_3 = 4 \cdot 3 + 3 \cdot 0 \cdot (-1) + 0 + 0 = 12$$

$$= 4 \cdot 3 = 12 = 1 \cdot (12)$$

$$\left(N(A_4) = \frac{4!}{2} = 3 \cdot 4 = 12 \right)$$

$\Rightarrow$ one of rep 3 is in rep 4.

$$\sum \text{Col}_4 \cdot \text{Col}_1 = 4 \cdot 1 + 0 + 4 + 4 = 12$$

$$= 1 \cdot (12)$$

$$\sum \text{Col}_4 \cdot \text{Col}_i = 4 \cdot 1 + 0 + 4w + 4w^* \quad // \quad 1 + w + w^* = 0$$

$$= 4(1 + w + w^*) = 0 = \sum \text{Col}_4 \cdot \text{Col}_{i'}$$

$\Rightarrow 4 \rightarrow 3+1$ the defining representation of A_4 reduces to 3 & 1 representations.

2 Show that the defining representation of S_n , when it breaks up, always contains the identity representation 1.

Defining representation in n -dimensional representation

For identity i to not be contained in defining representation we need

$$\sum_c n_c \chi^{*(i)}(c) \chi(c) = N(G) n_i = 0 \quad // \quad \text{Here } \chi(c) \text{ is reducible representation.}$$

It is then enough to show that $\sum_c n_c \chi^{*(I)}(c) \chi(c) \neq 0$

For I we have $\chi^{*(I)}(c) = 1$ for all c

$\Rightarrow \sum_c n_c \chi(c)$ n_c is positive definite

For defining representation $\chi(c)$ is positive definite because if

all elements are permuted we have $\text{tr } D(\text{all permuted}) = 0$

else we have $\text{tr } D(\text{not all permuted}) > 1$

becaus for example $(12)(34)$ is defined as

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \mathcal{D}((12)(34)) \Rightarrow \text{tr} \mathcal{D}((12)(34)) = 0$$

$$\Rightarrow \mathcal{D}((12)) = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \Rightarrow \text{tr} \mathcal{D}((12)) = 2 > 1$$

$\Rightarrow \chi(c)$ positive definiting

$$\Rightarrow \sum_c n_c \chi(c) = N(G) \cdot n_1 > 0$$

$\Rightarrow$ When refining rep of S_n breaks up it always contains the identity representation 1.

3 Determine the character table of S_4 , and check the various orthogonality theorems.

S_4 has 5 equivalent classes $\Rightarrow$ 5 representation:

$$\Rightarrow 1^2 + a^2 + b^2 + c^2 + d^2 = N(G) = 24$$

$$4! = 4 \cdot 3 \cdot 2 = 12 \cdot 2 = 24$$

$$\Rightarrow \text{The only sol is } 1^2 + 1^2 + 2^2 + 3^2 + 3^2 = 24$$

The character table looks like

S_n	n_c	c	1	$\bar{1}$	2	3	$\bar{3}$
	1	I	1	1	2	3	3
Z_1	3	(12)(34)	1	e			a
Z_2	8	(123)	1	f			b
Z_2	6	(12)	1	g			c
Z_3	6	(1234)	1	h			d

$$\chi(I) = d$$

We can determine the table with orthogonality theorem:

Starting with $C_I \cdot C_{\bar{3}} : 3 \cdot 1 + 3a + 8b + 6c + 6d = 0$

$C_I \cdot C_{\bar{1}} = 1 + 3e + 8f + 6g + 6h = 0 \Rightarrow |e| = |f| = |g| = |h| = 1$

we cannot have roots of 1 here since if we for example have

ω
 ω^2 with $1 + \omega + \omega^2 + \omega^2 + \omega^5 = 0$ this wouldn't satisfy orthogonality
 ω^3 between bases of different n_c 's.
 ω^5

$\Rightarrow e, f, g, h = \pm 1$ &

$1 + 3 + 8 - 6 - 6 = 0 \Rightarrow e = f = 1 \text{ \& } g = h = -1$

S_n	n_c	c	1	$\bar{1}$	2	3	$\bar{3}$
	1	I	1	1	2	3	3
Z_1	3	$(12)(34)$	1	1	i		a
Z_3	8	(123)	1	1	j		b
Z_2	6	(12)	1	-1	k		c
Z_3	6	(1234)	1	-1	l		d

now orthogonality says;

$C_I \cdot C_1 = 2 + 3i + 8j + 6k + 6l = 0$
 $C_{\bar{1}} \cdot C_2 = 2 + 3i + 8j - 6k - 6l = 0$
 $6k + 6l = -6(k+l) \Rightarrow k+l=0 \Rightarrow k=-l$

$\Rightarrow 2 + 3i + 8j = 0$
 $\Rightarrow$ only sol: $i=2, j=-1$
 since $\chi^{\text{irr}}(c) \leq \dim(V)$

Some argument is valid from orthogonality for 32 representations:

S_n	n_c	c	1	$\bar{1}$	2	3	$\bar{3}$
	1	I	1	1	2	3	3
Z_1	3	(12)(34)	1	1	2	d	a
Z_3	8	(123)	1	1	-1	e	b
Z_2	6	(12)	1	-1	k	f	c
Z_3	6	(1234)	1	-1	-k	-f	-c

$$C_I \cdot C_3 = 3 + 3d + 8e = 0 = C_{2'} \cdot C_3$$

$$\begin{aligned}
 C_3 \cdot C_2 &= 2 \cdot 3 + 2 \cdot 3 \cdot d - 8 \cdot e + 6kf + 6kf = 0 \\
 &= 6 + 6d - 8e + 12k \cdot f = 0 \quad || -8e = 3 + 3d \\
 &= 9 + 9d + 12k \cdot f = 0
 \end{aligned}$$

$$C_3 \cdot C_{1'} = 3 + 3d + 8e - 6f + 6f = 0 = 3 + 3d + 8e = 0$$

$$8 \cdot 3 = 24 > 3 + 3 \cdot 3 = 12$$

$$\Rightarrow \frac{e=0 \Rightarrow d=-1}{}$$

$$\Rightarrow C_3 \cdot C_2 = 12k \cdot f = 0$$

$$C_{\bar{3}} \cdot C_1 = 6 + 6a - 8b + 12k \cdot c = 0, \quad C_{\bar{3}} \cdot C_I = 3 + 3a + 8b = 0$$

$$\Rightarrow \underline{a=-1, \quad b=0}$$

$$C_{\bar{3}} \cdot C_2 = 9 + 9a + 12k \cdot c = 0$$

$$\begin{aligned}
 C_{\bar{3}} \cdot C_3 &= 9 + 3 \cdot d \cdot a + 8e \cdot b + 6fc + 6fc = 0 \\
 &= 9 + 3d \cdot a + 12fc + 8e \cdot b = 0
 \end{aligned}$$

$$\Rightarrow C_{\bar{3}} \cdot C_3 = 9 + 3 + 12f \cdot c = 0 \Rightarrow f \cdot c = -1$$

$$\text{Set } f=1, \quad c=-1$$

$$\Rightarrow k=0$$

7)

S_n	n_c	c	1	$\bar{1}$	2	3	$\bar{3}$
	1	I	1	1	2	3	3
Z_1	3	$(12)(34)$	1	1	2	-1	-1
Z_3	8	(123)	1	1	-1	0	0
Z_2	6	(12)	1	-1	0	1	-1
Z_3	6	(1234)	1	-1	0	-1	1

all of the column orthogonality satisfied

$$1 \cdot 3 + 3(1 \cdot 1) + 2 \cdot (3 \cdot 2) + 3^2 \cdot (-1) + 3^2 \cdot (-1) \\ = 3 + 3 + 12 - 9 - 9 = 18 - 18 = 0 \Rightarrow \text{row orthogonality}$$

4 Complete the table in (19).

D_4	n_c		1	$1'$	$1''$	$1'''$	2
	1	I	1	1	1	1	2
Z_2	1	$-I$	1	1	1	1	-2
Z_4	2	R, R^3	1	a	d	g	0
Z_2	2	r_x, r_y	1	b	e	h	0
Z_2	2	d_1, d_2	1	c	f	i	0

(19)

Col orthogonality:

$$C_1 \cdot C_{1'} = 0 = 1 \cdot 1 + 1 \cdot 1 + 2a + 2b + 2c = 0 \\ = 2(1 + a + b + c) = 0$$

$$C_1 \cdot C_{1''} = 2(1 + d + e + f) = 0$$

$$\Leftrightarrow C_{1'} \cdot C_{1''} = 2(1 + a \cdot d + b \cdot e + c \cdot f)$$

We can just guess $a = 1$, $b = -1$, $c = -1$

$$\Rightarrow 2(1 + a + b + c) = 0 \text{ is satisfied}$$

$$\Rightarrow C_{1'} \cdot C_{1''} = 2(1 + d - e - f) = 0$$

$$\Rightarrow 1 + d - e - f = 0 = 1 + d + e + f$$

$$\Rightarrow -e - f = e + f \Rightarrow e + f = 0$$

$$-(e + f) = e + f, \text{ if we chose } e = 1 \Rightarrow f = -1$$

We now have $\Rightarrow d = -1$

D_4	n_c		1	1'	1''	1'''	2
	1	I	1	1	1	1	2
Z_2	1	$-I$	1	1	1	1	-2
Z_4	2	R, R^3	1	1	1	g	0
Z_2	2	r_x, r_y	1	-1	1	h	0
Z_2	2	d_1, d_2	1	-1	-1	i	0

$$\Rightarrow 1 + g + h + i = 0 = \frac{1}{2} C_{1'''} \cdot C_1$$

$$C_{1'} \cdot C_{1''} = 2 + 2g - 2h - 2i = 0 \Rightarrow 1 + g - h - i = 0$$

$$C_{1''} \cdot C_{1'''} = 2 - 2g + 2h - 2i = 0 \Rightarrow 1 - g + h - i = 0$$

$$\Rightarrow g - h = h - g \Rightarrow 2g = 2h \Rightarrow g = h$$

if $g = -1$ then $i = 1$ is sol coz $i = -1$ doesn't give sol

$\Leftrightarrow g = i = h = 1$ is just trivial rep

D_4	n_c		1	1'	1''	1'''	2
	1	I	1	1	1	1	2
Z_2	1	$-I$	1	1	1	1	-2
Z_4	2	R, R^3	1	1	-1	-1	0
Z_2	2	r_x, r_y	1	-1	1	-1	0
Z_2	2	d_1, d_2	1	-1	-1	1	0

5 Work out the character table (21) for Q .

Q	n_c	1	1'	1''	1'''	2
	1	1				
Z_2	1	-1				
Z_4	2	$i, -i$				
Z_4	2	$j, -j$				
Z_4	2	$k, -k$				

There are 5 equivalence classes $\Rightarrow$ 5 irreducible representations

$$1 + a^2 + b^2 + c^2 + d^2 = N(G) = 8$$

$\Rightarrow a = b = c = 1$ & $d = 2$ are the representation dimensions.

The character table can now be constructed:

Q	n_c	1	1'	1''	1'''	2
	1	1				
Z_2	1	-1				
Z_4	2	$i, -i$				
Z_4	2	$j, -j$				
Z_4	2	$k, -k$				

$$C_1 \cdot C_{1'} = 1 + a + 2b + 2c + 2d = 0$$

$$C_{1'} \cdot C_{1''} = 1 + ae + 2bf + 2cg + 2dh = 0$$

$$C_1 \cdot C_{1''} = 1 + e + 2f + 2g + 2h = 0$$

So $1', 1'', 1'''$ can only have ± 1 because the corresponding cyclic group is Z_2 .

$\Rightarrow$ from $a, b, c, d = \pm 1$ we can choose

$$a = 1 = b, \quad c = d = -1$$

$$C_1 \cdot C_{1''} = 1 + e + 2f + 2g + 2h = 0$$

$$\text{then } C_{1'} \cdot C_{1''} = 1 + e + 2f - 2g - 2h = 0$$

all being 1 would work but that would be 1 rep & not 1'

$$\Rightarrow e = g = 1, \quad f = h = -1$$

⇒

Q	n_c		1	1'	1''	1'''	2
	1	1	1	1	1	1	2
z_2	1	-1	1	1	1	1 (m)	x
z_y	2	i, -i	1	1	-1	-1 (n)	y
z_z	2	j, -j	1	-1	1	-1 (p)	z
z_u	2	k, -k	1	-1	-1	1 (q)	u

$$C_{1'''} \cdot C_1 = 1 + m + 2n + 2p + 2q = 0$$

$1''' \neq \langle 1'', 1', 1 \rangle$ so only option is

$$q = 1, \quad p = -1, \quad n = -1, \quad m = 1$$

Now use row ortho to get x:

$$1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 + 1 \cdot 1 + 2 \cdot x = 0$$

$$4 + 2x = 0 \Rightarrow x = -2$$

$$C_1 \cdot C_2 = 1 \cdot 2 + 1 \cdot x + 2y + 2z + 2u = 0 \quad || x = -2$$

$$= 2y + 2z + 2u = 0$$

$$C_{1'} \cdot C_2 = 2y - 2z - 2u = 0$$

$$C_{1''} \cdot C_2 = -2y - 2z + 2u = 0$$

$$\Rightarrow 2u = -(2y + 2z) = + (2y + 2z) \Rightarrow u = 0$$

$$\Rightarrow 2y = 2z \quad \text{but} \quad 2y = -2z \Rightarrow y = z = 0$$

⇒

Q	n_c		l	l'	l''	l'''	2
	1	1	1	1	1	1	2
Z_2	1	-1	1	1	1	1	-2
Z_4	2	$i, -i$	1	1	-1	-1	0
Z_4	2	$j, -j$	1	-1	1	-1	0
Z_4	2	$k, -k$	1	-1	-1	1	0

Same as D_n lmao.

6.

6 For $n \geq 5$, any two 3-cycles in A_n are equivalent. Why is the restriction $n \geq 5$ needed?

Γ_n $A_n \leftarrow A_3$ (123) & (132) aren't equivalent

let's see if for $n \geq 5$ they are

let $s = (245)t$ where t not in A_n but $t \in S_n$

$S \in A_n$ so we can for example use $S(123)S^{-1}$ to show we can get to any other 3-cycle:

$$S^{-1}(123)S = t^{-1}(54)(123)(45)t = t^{-1}(123)t$$

$t \in S_n$ & $t \notin A_n$ & in S_n every 3-cycle is equivalent via

$t^{-1}(123)t \Rightarrow$ same happens in A_n for $n \geq 5$ too

we need $n \geq 5$ because for $n < 5$ there aren't enough elements

i.e. there is no (45) so that we would have $s = (45)t$

to transform (123) into other 3-cycles.

7 Work out the character table for $D_5 = C_{5v}$, the invariance group of the pentagon. You will find, perhaps surprisingly (perhaps not), that the character table for D_5 is smaller than the character table for D_4 .

$$D_5 = \{ R, r \mid R^5 = I, r^2 = I, Rr = rR^{-1} \}$$

$$\Rightarrow D_5 = \{ I, R, R^2, R^3, R^4, r, rR, rR^2, rR^3, rR^4 \}$$

$\xleftarrow{\text{ant}} \xrightarrow{\text{ant}}$ $\uparrow$
 anti anti

I is in class of its own

$$rRr = R^{-1} \Rightarrow R \text{ \& } R^{-1} \text{ in class}$$

$$RR^{-1} = I \Rightarrow R^{-1} = R^4$$

$$R^2(R^2)^{-1} = I \Rightarrow (R^2)^{-1} = R^3$$

$$\underline{rRr = R^4}, \quad \underline{rR^2r = rRrR^{-1} = (R^{-1})^2 = R^3}$$

$$r^{-1} = r \Rightarrow r^{-1}rRr = Rr = rR^{-1} = rR^3$$

$$\Rightarrow rR \text{ \& } rR^3 \text{ equiv}$$

$$r^{-1}rR^2r = r^{-1}R^3 = rR^3 \Rightarrow rR^3 \text{ equiv with } rR^2$$

$$\text{Class: } (R^2)^{-1}R R^2 = R^3 R R^2 = R \quad \text{no equivs with } R^2, R \text{ done}$$

$$\text{Sim: } (rR)^{-1}R(rR) = rR R rR = R^{-1} = R^4$$

$$R^{-1}(rR)R = R^{-1}rR^2 = rR R^2 = rR^3$$

$$rR \text{ equiv with } rR^3$$

$$R^2 (rR) R^2 = R^3 r R^3 = r R^{-3} R^3 = r \quad , \quad \begin{array}{l} rR \sim rR^3 \\ rR \sim r \\ rR \sim rR^2 \end{array}$$

$$R^3 (rR) R^3 = R^2 r R^4 = r R^2 \quad rR \sim rR^4$$

$$R^{-1} (rR) R^{-1} = R r = r R^{-1} = r R^4$$

$$(rR) R (rR) = r R r R^{-1} R = r^2 R^{-1} = r^2 R^4 \Rightarrow R \sim R^4$$

$$(rR) R^2 rR = r R^3 r R = r^2 R^{-3} R = R^{-2} = R^5 \quad \text{ie} \quad R^2 \sim R^5$$

$$(rR^4) R^2 (rR^4) = r R r R^4 = R^3$$

$$r R^3 R^2 r R^3 = r^2 R^{-5} R^3 = R^3$$

$$r R^2 R^2 r R^2 = r^2 R^{-4} R^2 = R^3$$

$$R^2, R^3 \text{ commute with each other but do note } R^1 = R^{-3} \leftarrow R^{-1} = R^4$$

$$(rR) R (rR) = r^2 R^{-1} = R^4 \Rightarrow R \sim R^4$$

Equiv classes there are 4 : $I, C = \{R, R^4\}, C = \{r, rR, rR^2, rR^3, rR^4\}, C = \{R^2, R^3\}$

$\Rightarrow$ 3 representations exist in the table below like following:

$$\Rightarrow I^2 + a^2 + b^2 + c^2 = 10$$

$$\Rightarrow a = b = 2, \quad c = 1$$

D_i	n_i		1	1'	2	2'
1	I		1	1	2	2
2	R, R^4		1	a	d	q
2	R^2, R^3		1	b	e	h
5	$rR, rR^2, \dots$		1	c	f	l

Orthogonality: $C_1 \cdot C_1 = 0 \Rightarrow 1^2 + 2a + 2b + 5c = 0$

$\Rightarrow 1^2 + 2a + 2b + 5c = 0 \quad || \quad a=b=1, c=-1$
the only solution

$C_1 \cdot C_2 = 2 + 2d + 2e + 5f = 0$

$C_2 \cdot C_1 = 2 + 2ad + 2be + 5cf = 2 + 2d + 2e - 5f = 0$

$\Rightarrow 5f = -5f \Rightarrow f = 0$

$\Rightarrow 2(1 + d + e) = 0$

options: $\left\{ \begin{array}{l|l} d=-2, c=1 & d=-1, e=0 \\ d=1, e=-2 & e=-1, d=0 \end{array} \right.$

one of them gives 2 and another gives 2'

$\Rightarrow 2: d=-1, e=0 \quad \& \quad 2': d=1, e=-2$

$\Rightarrow$

D_s	n_c		1	1'	2	2'
	1	I	1	1	2	2
	2	$R_1 R_1'$	1	1	-1	1
	2	R^2, R^3	1	1	0	-2
	5	$rR_1, rR_2, \dots$	1	-1	0	0

$C_1' \cdot C_1 = 2 + 2 - 2 \cdot 2 = 0$